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Assignment-5
Information Theory and Coding

Q1. A convolutional encoder has constraint length 3. A student says number of states = 3. Verify and correct.
Ans. Number of states = 2 (K-1)
Here K = 3
States = 2 (3-1) = 2 2 = 4
# Conclusion:
The student is incorrect
Correct number of states = 4

Q2. Code rate is 1/2, but output sequence length is not exactly double input. Explain why
Ans. Code rate 1/2 means ideally output = 2 × input bits

However, termination (tail bits) are added to return encoder to zero state
These extra bits increase output length
Conclusion:
Output is slightly more than double due to termination bits

Q3. Two encoders: Encoder A: K=3, Encoder B: K=5. Which is better?
Ans. Larger constraint length → more memory → better error correction
Encoder B (K=5) has more states and larger free distance
Conclusion:
Encoder B has better error performance
Trade-off: higher complexity

Q4. A received sequence perfectly matches a valid path in trellis. Can error still exist?
Ans. Yes, possible
Decoder selects most likely path, not guaranteed correct path

Noise may push wrong path to appear valid
Conclusion:
Error can still exist even if path matches trellis

Q5. Switching from hard decision to soft decision decoding. What improvement?
Ans. Soft decision uses more information (probability levels)
Typically gives 2 dB coding gain improvement
Conclusion:
Expect around 2 dB performance improvement

Q6. Convolutional encoder (2,1,3), generator (7,5), input 1011

Ans. Step 1: Convert generators to binary
$7 \longrightarrow 111$
$5 \longrightarrow 101$

Step 2: Initial state = 00
Step 3: Encoding process
- Input 1: shift register = 100
- Output: (111˙100 = 1), (101˙100 = 1) → 11

- Input 0: shift register = 010
Output: (111˙010 = 1), (101˙010 = 0) → 10

- Input 1: shift register = 101
Output: (111˙101 = 0), (101˙101 = 0) → 00

- Input 1: shift register = 110
Output: (111˙110 = 0), (101˙110 = 1) → 01

- Encoded sequence: 11 10 00 01

- Step 4: Partial trellis (states: 00, 01, 10, 11)
- From each state, input 0 or 1 leads to

next state with outputs

Example:

- 00 ⟶ input 1 ⟶ 10 (output 11)
- 10 ⟶ input 0 ⟶ 01 (output 10)
- 01 ⟶ input 1 ⟶ 10 (output 00)

Step 5: Competing paths
Different paths may produce similar output sequences
Example:
Path1: 00⟶10⟶01⟶10 (correct path)
Path2: 00⟶00⟶00⟶00 (all-zero path)
These compete in Viterbi decoding

Conclusion:

- Encoded output = 111000001
- Trellis shows multiple competing paths used in decoding